

Geometric Origin of Nucleon Binding and a Programme for Beta Decay in HQIV

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Abstract

We develop a self-contained geometric account of nucleon binding within the Horizon-Quantized Information Vacuum (HQIV) framework, built on a 3D discrete null lattice with an octonionic eightfold mode count. Cumulative simplex counting yields a shell-wise curvature-imprint ratio identically equal to $\alpha = 3/5$; the informational monogamy complement is then $\gamma = 1 - \alpha = 2/5$. We record the synchronous-comoving lapse $N = 1 + \Phi + \phi t$ (natural units), the identification of the horizon auxiliary ϕ across the emergent gauge sector and an effective Friedmann coupling $G_{\text{eff}}(\phi) = \phi^\alpha$, and a concrete *divisor* prescription that renormalizes nucleon meta-harmonic masses on the lock-in shell when $N > 1$. Elementary inequalities show the lapse-corrected readout is strictly below the uncorrected mass in that regime. Beta decay is proposed as a curvature-balance transition coupled to *skew* time-phase alignment on the octonionic carrier; explicit lifetimes and a full nuclear potential are not derived here. We close with a short comparison to Standard Model parameter counts and with an appendix that maps each major claim to an independent machine-checked certificate in a public repository—an audit layer that does not substitute for the in-text arguments.

Update (v2). Since the first working copy, the repository has accumulated *computed witnesses* on the same discrete spine: free-neutron integrated clock time $\tau_n \approx 880$ s at 300 K; light-isotope mass readouts at mean $\approx 0.12\%$ vs. tabulated comparison masses; tritium β^- width matched to reference after a two-ledger valley prescription (symmetric cluster binding for masses, interior valence well for widths); outside temperature $T(\xi)$ and a gravity-stack modulator at sub-ppm on τ_n but percent-level at BBN temperatures; local curvature neutrino opacity on free-branch weak width; and a condensed-phase pipeline (`hqiv_lab`) that *derives* bulk allotropes from molecular bonds rather than tabulated crystal names, yielding ρ_{curv} and melt readouts for H_2O ice Ih at ≈ 272 K. Sections 5–7 record these results at the level of definitions and order-of-magnitude numbers; full tables and reproducer scripts are in the companion note `hqiv_nucleon_binding_from_composite_trace.tex`.

1 Discrete null lattice and the imprint identities

1.1 Mode counts

Fix a non-negative shell index m . The positive quadrant slice $\{(x, y, z) \in \mathbb{Z}_{\geq 0}^3 : x + y + z = m\}$ has $\binom{m+2}{2}$ integer points (stars-and-bars). Following HQIV’s octonionic carrier bookkeeping, attach a factor of eight to obtain the *available* horizon modes at shell m ,

$$A(m) = 8 \binom{m+2}{2} = 4(m+1)(m+2). \quad (1)$$

Write $N_{\text{simp}}(m) := (m+1)(m+2)$ for the *stars-and-bars numerator* attached to shell m in the HQIV convention (equal to twice a binomial coefficient, $2 \binom{m+2}{2}$), if one insists on the classical

$\binom{m+2}{2}$ point count). The cumulative count up to n is

$$C(n) := \sum_{m=0}^n N_{\text{simp}}(m). \quad (2)$$

Lemma 1 (Cubic cumulant / hockey-stick). *For every $n \in \mathbb{N}$,*

$$3C(n) = (n+1)(n+2)(n+3).$$

$$\text{Equivalently } C(n) = \frac{(n+1)(n+2)(n+3)}{3}.$$

Proof. Induction on n . For $n=0$, $C(0) = N_{\text{simp}}(0) = 2$ and $(0+1)(0+2)(0+3) = 6 = 3C(0)$. Assume $3C(n) = (n+1)(n+2)(n+3)$. By definition $C(n+1) = C(n) + N_{\text{simp}}(n+1)$ with $N_{\text{simp}}(n+1) = (n+2)(n+3)$. Thus

$$\begin{aligned} 3C(n+1) &= 3C(n) + 3(n+2)(n+3) \\ &= (n+1)(n+2)(n+3) + 3(n+2)(n+3) \\ &= (n+2)(n+3)((n+1)+3) \\ &= (n+2)(n+3)(n+4), \end{aligned}$$

which is the desired identity with n replaced by $n+1$. □

Definition 2 (Curvature-imprint ratio). For $n \in \mathbb{N}$ define

$$r(n) := \frac{(n+1)(n+2)(n+3)}{5C(n)}.$$

Theorem 3 (Forced imprint exponent). *For every $n \in \mathbb{N}$, $r(n) = 3/5$. Define $\alpha := 3/5$ and $\gamma := 1 - \alpha = 2/5$.*

Proof. By Lemma 1, $(n+1)(n+2)(n+3) = 3C(n)$, hence $r(n) = 3C(n)/(5C(n)) = 3/5$ because $C(n) > 0$ for all n . □

Remark 4. The rational $\alpha = 3/5$ is therefore not a tunable exponent within this discrete normalization: it is the unique value of the ratio (2) once (2) is adopted. The companion $\gamma = 2/5$ is the unique affine complement on the unit split $\alpha + \gamma = 1$.

1.2 Temperature ladder and auxiliary field

Definition 5 (Shell temperature and auxiliary ϕ). Set $T(m) := 1/(m+1)$ in natural units and $S(m) := (m+1)(m+2)$. The HQIV homogeneous auxiliary field at shell m is

$$\phi(m) := \frac{2}{T(m)} = 2(m+1).$$

1.3 Electroweak scale witnesses (algebraic)

The geometric vacuum expectation used in the HQIV electroweak package is assembled from the lock-in shell index m_{lockin} , the ladder T , the surface factor S , and the monogamy complement $(1+\gamma)$; we do not reproduce the full triality-normalized coupling algebra in this note. Specializing the closed rational outputs recorded in the development gives the numerical identities

$$M_W^{\text{derived}} = \frac{392}{5}, \quad m_H^{\text{derived}} = \frac{588}{5} \quad (3)$$

in the same mass units used for those lemmas, together with a companion neutral-boson relation tying M_Z^{derived} to M_W^{derived} and γ .

2 Three naturals, one real, and skew time-phase alignment

2.1 Synthesis theorem (quoted package)

Under the *exact* quadratic shell law $A(m) = 4(m+1)(m+2)$, transverse null-mode counting is modeled by three independent nonnegative integers $(x, y, z) \in \mathbb{N}_0^3$ with $x + y + z = m$ (the “three naturals”), while causal height along maximal chains supplies an ordered timelike axis modeled by a single copy of $\mathbb{R}$ (the “one real”). The Fano-plane / octonion internal structure is then tied to the projective geometry of $\mathbb{F}_2^3$, and the imprint pair $(\alpha, \gamma) = (3/5, 2/5)$ is the $d = 3$ row of the standard HQIV unit-split family. A fully self-contained statement and proof sketch of this **Three Natural + one Real** package appears in the companion HQIV manuscript on 3D causal growth and octonionic gauge (Theorem “Three Natural + one Real under the HQIV shell hypothesis” there); the present note uses only the counting input (1) and Theorem 3.

2.2 Jobs of ϕ and N , and a fourth skew job

Already in the HQVM chart one assigns three roles to the auxiliary field ϕ and the lapse $N = 1 + \Phi + \phi t$ (with $c = 1$): (i) temperature / shell ladder ($T(m) = 1/(m+1)$, $\phi(m) = 2(m+1)$); (ii) hypercharge-like U(1) phase data via the octonionic scalar slot and the distinguished phase-lift generator $\Delta \in \mathfrak{so}(8)$ acting on $\mathbb{R}^8$; (iii) superposition bookkeeping on the $\mathbb{R}^8$ carrier. We now add a fourth, explicitly *geometric* role: **skew alignment** of the internal phase with the real timelike axis selected by the lapse.

The generator Δ is antisymmetric (skew-adjoint) in the (e_1, e_7) plane in the standard octonion matrix model; octonion multiplication supplies non-associativity and a natural associator 3-form that can be read as an algebraic “skew” coupling between the three transverse null directions and the ordered time axis. The proposal is that nuclear binding and decay channels are selected not only by energy and charge, but by whether a given resonance product can *close* curvature and *realign* this skew factor inside a deep potential well.

Axiom 6 (Skew time-phase alignment). Fix a skew-adjoint endomorphism $\Delta \in \mathfrak{so}(8)$ representing the phase-lift direction on $\mathbb{R}^8 \cong \mathbb{O}$. Let $s(t) > 0$ denote the positive scalar factor assembled from the horizon auxiliary and lapse readout along the comoving slice (reducing to ϕt in the homogeneous sector up to normalization choices), and let $\theta : \mathbb{R} \rightarrow \mathbb{R}$ be an angular coordinate. The internal time-phase factor is modeled by

$$\Psi(t) := s(t) \exp(\theta(t) \Delta) \in \mathbb{R}_{>0} \times \mathrm{SO}(8), \quad (4)$$

with $\dot{\theta}$ proportional to the cumulative lapse increment along the chosen comoving observer in normalized units.

Remark 7 (Status). Equation (4) packages skew alignment as a modeling axiom on the $\mathrm{SO}(8)$ carrier. What is *proved* in Lean: shell identities (Theorem 3); skew-adjointness of the concrete Δ matrix (`phaseLiftDelta_antisymm`); ledger separation (strong overlap vs weak β , `NeutronDecayWidthLedger`); bottle/beam method structure (`temperature_ppm_insufficient_for_bottle_beam_apparent_tau_gt_bottle_apparent_tau`). Variational wiring of (4) into nuclear wells remains a quantitative extension of `TrapWeakBridgeDecoherenceSlot`.

3 Lapse, gravity coupling, and nucleon divisor readouts

3.1 Synchronous lapse and shared ϕ

Definition 8 (ADM lapse in the HQVM chart). Let Φ be the Newtonian potential, ϕ the horizon auxiliary, and t coordinate time (natural units, $c = 1$). Define the lapse

$$N(\Phi, \phi, t) := 1 + \Phi + \phi t. \quad (5)$$

Definition 9 (Effective gravitational coupling). For $\phi \geq 0$ set $G_{\text{eff}}(\phi) := \phi^\alpha$ with $\alpha = 3/5$ from Theorem 3.

Remark 10 (Compatibility package). The HQIV construction posits that the *same* real scalar ϕ enters the lapse time-angle term ϕt , an emergent O-type Maxwell sector, and the Friedmann-side coupling G_{eff} . The precise variational derivation of Einstein equations from matter is *not* claimed here; what is used below is the algebraic synchronization of symbols across sectors.

3.2 Lock-in shell and binding readout

Definition 11 (Lock-in index). $m_{\text{lockin}} \in \mathbb{N}$ denotes the discrete shell at which QCD-scale structure is aligned with horizon lock-in in the HQIV ladder. It is treated as a fixed structural index in the same sense as a renormalization scale in ordinary field theory: downstream masses are *evaluated* there, not solved for from (1) alone.

Let $B_\star > 0$ denote the shared binding energy (in mass units) extracted from the composite-trace functional at m_{lockin} . Let $M_u, M_d > 0$ be dressed constituent masses for the light quarks in the same units, assembled into proton and neutron *constituent* totals

$$M_p^{\text{const}} := 2M_u + M_d, \quad M_n^{\text{const}} := M_u + 2M_d.$$

Define the *raw* meta-harmonic nucleon masses by subtraction of the shared binding,

$$M_p^{\text{raw}} := M_p^{\text{const}} - B_\star, \quad M_n^{\text{raw}} := M_n^{\text{const}} - B_\star. \quad (6)$$

On the lock-in shell set $\phi_\star := \phi(m_{\text{lockin}})$ and define the restricted lapse $N_\star(\Phi, t) := N(\Phi, \phi_\star, t)$.

Proposition 12 (Lapse divisor correction). *Define corrected masses by*

$$M_p^{\text{corr}}(\Phi, t) := \frac{M_p^{\text{raw}}}{N_\star(\Phi, t)}, \quad M_n^{\text{corr}}(\Phi, t) := \frac{M_n^{\text{raw}}}{N_\star(\Phi, t)}.$$

If $N_\star(\Phi, t) > 0$, the prescriptions are well-defined.

Proof. Immediate from (5) and division on $\mathbb{R}$. □

Lemma 13 (Positivity of $\phi_\star$). *For every shell index m , $\phi(m) > 0$. In particular $\phi_\star > 0$.*

Proof. By Definition 5, $\phi(m) = 2(m+1) > 0$. □

Proposition 14 (Weak-field forward-time regime). *If $1 + \Phi > 0$, $\phi = \phi_\star$, and $t \geq 0$, then $N(\Phi, \phi, t) > 0$. If in addition $\Phi \geq 0$ and $t > 0$, then $N(\Phi, \phi, t) > 1$.*

Proof. From (5), $N = 1 + \Phi + \phi t \geq 1 + \Phi > 0$ when $t \geq 0$ and $\phi > 0$ (Lemma 13). If $\Phi \geq 0$ and $t > 0$, then $\phi t > 0$ and hence $N > 1 + \Phi \geq 1$, in fact $N > 1$ because $\phi t > 0$. □

Corollary 15 (Mass readout strictly decreases when $N > 1$). *If $M_p^{\text{raw}} > 0$ and $N_\star(\Phi, t) > 1$, then $M_p^{\text{corr}}(\Phi, t) < M_p^{\text{raw}}$. The same statement holds with M_n in place of M_p .*

Proof. For $N > 1$ and $M > 0$, $M/N < M$ is equivalent to $M < MN$, i.e. to $1 < N$ after cancelling M . □

3.3 Isospin splitting (model input vs. data)

The HQIV nucleon package separates a fixed hypercharge sign contribution (normalized to a 1 MeV bookkeeping gap in the formal witness) from a residual mass difference driven by the dressed light-quark average and detuning dress terms. The observed neutron–proton mass difference ≈ 1.293 MeV is quoted only as a phenomenological benchmark: matching it quantitatively is not asserted in this working copy.

Remark 16 (PDG gap). Current meta-harmonic nucleon outputs in the computational pipeline sit several percent above PDG masses; closing that gap is stated as an open strong-sector and dressing problem, not as a defect of Theorem 3.

4 A programme for beta decay

Conjecture 17 (Curvature and skew: β transitions). *Outside compact nuclear environments, a free neutron occupies a branch of the curvature ledger with sub-lock-in normalized curvature (Ω_k -type ratios below unity relative to the calibration shell) and with **skew misalignment** of the internal phase (4) relative to the real timelike axis. Embedding in a nuclear potential well together with lapse reinforcement is conjectured to supply both sufficient depth and a triality-compatible skew realignment, closing curvature only for resonance products whose $\text{Spin}(8)$ triality channel matches the local skew configuration (charge and magnetic moment are partial selectors; skew completes the orientation). Conversely, proton-rich configurations whose wells overshoot in skew-depth are conjectured to reopen the imbalance and relax via β^+ or electron-capture channels—the inverse geometric channel to neutron stabilization.*

Remark 18 (Scope). Conjecture 17 is deliberately schematic: no Fermi theory, no W propagator, and no partial width is derived. The role of the weak sector in HQIV is instead represented by structural reductions of an octonionic weak–Higgs scaffold onto an abelian Maxwell-type backbone, together with triality bookkeeping that enforces a neutral CP-tilt average and a constructive spin–statistics package compatible with horizon-limited continuum scaffolds. Because Δ acts on the full $\mathbb{R}^8$ carrier, the neutron’s nonzero magnetic moment participates in skew/triality selection even though the neutron is electrically neutral at the $U(1)$ hypercharge level.

5 Computed witnesses on the discrete spine

Conjecture 17 is schematic; the repository nonetheless exports *witness readouts* that fix numerical slots on the same (α, γ) ladder without $\overline{\text{MS}}$ current-quark inputs or fitted half-lives. The following is a summary of the current pipeline (reproducer commands in Appendix A).

5.1 Two-ledger β width prescription

Mass readouts use symmetric cluster binding B_{cluster}/A on the composite trace at lock-in. *Width* readouts use a separate interior valence well: for $A = 1$ the free neutron sees the full exterior weak channel; for $A > 1$ the interior ledger is shielded by bonded closure. Naive use of B_{cluster}/A in the width slot over-suppressed tritium; the interior well restores β^- width agreement to reference order without a tritium-specific fit.

5.2 Free neutron and light panel

At laboratory temperature 300 K with the electroweak G_F slot from the dynamic β chart, the integrated clock time is

$$\tau_n \approx 880 \text{ s}, \quad \Gamma_{\beta^-} = \ln 2 / \tau_n \approx 7.9 \times 10^{-4} \text{ s}^{-1},$$

consistent with bottle/beam central values (878–888 s spread quoted as comparison only). Light-isotope masses from the same network land at mean $\approx 0.12\%$ relative to tabulated comparison entries; tritium half-life matches reference after the two-ledger width rule.

5.3 Molecular embedding (tritium gas and water)

Laboratory tritium is rarely a bare nucleus. The bond-state network supplies an *outside* contact sum for molecules containing the decaying centre; this folds through the same outside G_{eff} modulator as the gravity stack. T_2 gas at 300 K adds $\mathcal{O}(10 \text{ ppm})$ to the tritium half-life—the same *kind* of shift as galactic-scale gravity on τ_n ($\sim 0.15 \text{ ppm}$ for the full Earth–Sun–Galaxy stack), confirming that outside geometry is a small perturbation on free-neutron decay in the lab but not identically zero.

5.4 Bottle versus beam (Lean-backed structure)

The $\sim 1.2\%$ split is real; HQIV does not fit method knobs to 878 s / 888 s. Lean proves: (i) both methods read weak Ledger III, not strong Ledger I; (ii) 4 K vs 300 K outside support is $\sim 10^3$ ppm, far below the split; (iii) central $\tau_n \approx 880$ s lies between the representative references; (iv) $\tau_{\text{beam}} > \tau_{\text{bottle}}$ once trap width dressing exceeds beam dressing. Open: numeric discharge of trap-bridge decoherence and wall-loss channels (`TrapWeakBridgeDecoherenceSlot`).

6 Outside temperature, gravity, and local neutrino opacity

6.1 Outside release at $T(\xi)$

Nucleon own-binding and outside caustics are modulated by an outside release factor evaluated at $T(\xi) = T_{\text{P1}}/\xi$. At lock-in the modulator normalises to unity; at BBN-relevant ξ the outside support weakens to ≈ 0.785 —percent-level coupling to light-element abundances (lithium problem regime), distinct from the sub-ppm laboratory neutron clock.

6.2 Gravity stack

A dimensionless gravity-stack factor ε_g (Earth, Sun, Galaxy layers from the flyby scaffold) enters the outside G_{eff} modulator on free branches. At 300 K the full stack shifts τ_n by ~ 0.15 ppm: measurable in principle, blind at current bottle precision.

6.3 Local curvature neutrino opacity

On the free-neutron branch the weak width carries a multiplicative catalysis from local curvature neutrino opacity:

$$\sigma_{\nu,\text{eff}} \approx (140 \text{ MeV})^4 B_{\text{curv}} f_g \quad (\text{barn scale, OOM witness}), \quad \delta_w \approx 7.3\% \text{ at lock-in with } \pm\gamma/5 \text{ envelope.}$$

Bonded clusters ($A > 1$) shield interior decays; the opacity slot is active primarily on free and weakly embedded branches. The BBN integrator exports `local_curvature_neutrino_width` on the same spine.

7 Condensed-phase curvature outside the nuclear well

The outside-contact primitive used for molecular embedding extends to *bulk* condensed matter without Avogadro-scale atom counting. Inputs are molecular specs (formula weight, covalent lengths and angles, intermolecular ξ , cohesive energies). **Allotropes are derived**, not chosen from a static crystal-name table.

7.1 HQIV chem & materials lab (`hqiv_lab`)

The `hqiv_lab` package implements:

1. VSEPR motif inference (e.g. tetrahedral H-bond for H_2O);
2. packing templates (Ih, Ic, fcc, amorphous, ...);
3. unit cell (a, b, c, Z) from contact distance $\sim 2r_{\text{bond}}(1 + \alpha)$;
4. ranking at (T, P) on the thermodynamic spine.

CLI: `hqiv_lab H2O`. Lean structural mirror: `Hqiv/QuantumChemistry/PhaseAllotropeDerivation.lean`.

7.2 Curvature density and melt

Mass density $\rho_{\text{mass}} = ZM/(N_A V_{\text{cell}})$ and

$$\rho_{\text{curv}} = \text{clamp}_{[0,1]} \left(\frac{\rho_{\text{mass}}}{\rho_{\text{liquid,ref}}} \right)$$

feed the homogeneous second-order budget $B_{\text{hom}}(\xi, \rho) = 1 + \rho_{\text{curv}}(B_{\text{curv}}(\xi) - 1)$. For derived H₂O ice Ih at 273 K, 1 atm:

Quantity	Witness
Preferred allotrope	Ih ($\succ$ Ic $\succ$ amorphous)
ρ_{mass}	$\approx 0.85 \text{ g cm}^{-3}$
Cell (a, c)	$\approx (5.61, 5.18) \text{ \AA}, Z = 4$
ρ_{curv}	≈ 0.85
T_{sl} (bulk)	$\approx 272 \text{ K}$

Numbers follow bond geometry and $\alpha = 3/5$ opening; they are not fitted to NIST hexagonal cell constants ($a \approx 4.515 \text{ \AA}$, $\rho \approx 0.92 \text{ g cm}^{-3}$), which may serve as optional calibration witnesses only.

7.3 Material response on the same spine

Once ρ_{curv} and allotrope orientation are fixed, downstream readouts include refractive index n (Clausius–Mossotti from binding-softness polarizability), thermal conductivity k_{th} , molar heat capacity $C_{p,\text{mol}}$, latent heat L_{fusion} (structural slot), liquid viscosity η (Eyring barrier from E_{melt}), and ice-Ih birefringence Δn from hexagonal c/a split. Witness at 273 K for derived Ih: $n \approx 1.24$, $C_{p,\text{mol}} \approx 119 \text{ J mol}^{-1} \text{ K}^{-1}$, $k_{\text{th}} \approx 1.2 \text{ W m}^{-1} \text{ K}^{-1}$, $\Delta n \approx 4 \times 10^{-4}$; liquid water $n \approx 1.33$, $C_{p,\text{mol}} \approx 74 \text{ J mol}^{-1} \text{ K}^{-1}$. Absolute L_{fusion} vs. tabulated 333 kJ mol^{-1} is *not* claimed—only the structural melt slot is wired.

8 Large-scale signatures

Remark 19 (Low multipoles). If the lapse-selected time slicing couples coherently to the longest-wavelength modes of the background, one expects modified low- ℓ statistics in polarization and temperature (sometimes colloquially summarized as “axis-of-evil”-type alignments). Any such prediction must eventually pass through a full Einstein–Boltzmann transport implementation; that layer is not part of the present note.

Separately, a purely algebraic *theory hook* relates a birefringence angle defined from $(\alpha, m_{\text{lockin}})$ to a redshift factor $(1+z) = (m_{\text{lockin}} + 1)^\alpha$ at a chosen normalization $\kappa_\beta = 1$. This identity is a consistency display, not a claim that the cosmic birefringence angle has been measured to equal that value.

9 Comparison to the Standard Model

The electroweak sector of the Standard Model introduces an independent non-abelian weak gauge structure and, in textbook counts, on the order of twenty-six freely fitted parameters. The HQIV presentation here reduces a portion of that freedom to discrete counting, a single imprint exponent forced by Lemma 1, and explicit structural pins (lock-in index, quark-sector anchors) whose remaining arbitrariness is acknowledged rather than hidden.

Acknowledgements

The author acknowledges AI-assisted drafting and code navigation; mathematical claims in the main text are intended to be verifiable by hand from the definitions given.

Further reading (HQIV programme). [2, 1]

A Formal audit layer (optional)

The repository <https://github.com/disregardfiat/hqiv-lean> contains a Lean 4 formalization used as an *independent check* of the algebraic claims above: Theorem 3 is the statement `latticeAlphaRatio_eq_alpha` specialized to the definition of α ; Lemma 13 is `phi_of_shell_pos`; Proposition 14 re-packages `HQVM_lapse_pos` and `HQVM_lapse_gt_one`; Corollary 15 is `protonMassFromMetaHarmonics`. The divisor law Proposition 12 is `protonMassFromMetaHarmonics_lapseCorrected_eq_raw_div_lapse`. Skew-adjointness of the concrete phase-lift matrix for Δ is `phaseLiftDelta_antisymm` in `Hqiv/Algebra/PhaseLiftDelta.lean` (re-exported through the $\text{SO}(8)$ / QFT feed modules). For patch-local digital evolution, `Hqiv/QuantumComputing/HamiltonianToGateMapping.lean` formalizes sequential gate schedules with per-tick η_ϕ -CKW certificates (`SequentialHQIVEvolution`), dense execution `runSequentialEvolution` (aligned with `DiscreteSchrodinger`), and an OS-Horacle sparse trajectory `runSparseSequentialOSH`; these layers audit one-step-at-a-time Hamiltonian *proxies* on the discrete cone without changing the shell-counting inputs used above.

Witness modules (Sections 5–7). Dynamic β chart and stability readouts: `Hqiv/Physics/DynamicBetaIso`, `Hqiv/Physics/DynamicIsotopeStability.lean` (`betaWidthLedgerQualified`, `EMTippingQualified`). Neutron lifetime methods (bottle/beam): `Hqiv/Physics/NeutronLifetimeMethod.lean`. Outside temperature and G_{eff} support: `Hqiv/Physics/NuclearOutsideTemperatureDynamics.lean`. Phase geometry and material response: `Hqiv/QuantumChemistry/PhaseGeometryDensity.lean`, `Hqiv/QuantumChemistry/PhaseMaterialResponse.lean`, `Hqiv/QuantumChemistry/PhaseAllotropeDerivative` (motif `h2o_has_ice_1h_label`). Homogeneous curvature second order: `Hqiv/Physics/HomogeneousCurvature`.

Python reproducers. From repository root:

```
python3 scripts/hqiv_isotope_pdg_benchmark.py --qualify-em-tipping
python3 scripts/hqiv_isotope_stability_half-life.py --A 1 --Z 0 \
  --qualify-em-tipping --lab-temperature-K 300 --gravity-stack full
pip install -e . && hqiv-lab H2O
python3 scripts/hqiv_phase_geometry_density.py H2O
PYTHONPATH=scripts python3 scripts/test_hqiv_phase_material_response.py
```

Full witness tables and claim map: `hqiv_nucleon_binding_from_composite_trace.tex` (v6).

Axiom 6 is **not** a Lean theorem: it records the modeling target coupling Δ to the lapse-selected slice. No result in this appendix substitutes for the proofs in Sections 1–2: it is an audit trail for readers who wish to replay the certificates mechanically.

References

References

- [1] K. Brodie. Derivation of quantised inertia from Jacobson thermodynamics. Zenodo, 2026. Backward-hemisphere overlap; factor $1/6$ and thermodynamic coefficient γ .

- [2] Steven Jr Ettinger. Horizon-quantized informational vacuum (hqiv): A unified framework from causal horizon monogamy and discrete null-lattice combinatorics. Zenodo preprint, 2026. Version v2.